University of Amsterdam

Individual Modelling Assignment

Pumping Up the Heat

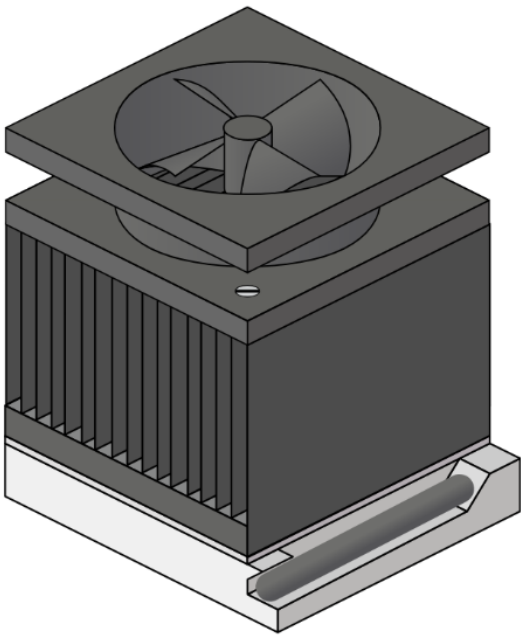
How Tech Costs, Tech Efficiency, and Carbon Pricing Shape Industrial Heat Electrification and Decarbonisation in the Netherlands

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Course:
Modelling for Transitions



(Pixabay 2026)

Introduction

The 2015 Paris Agreement, the Sustainable Development Goals, and the COVID-19 pandemic created new momentum for industrial decarbonization, with electrification emerging as a major mitigation strategy (IPCC WG III Report 2022). Heating and cooling make up about 50% of all energy consumed in the EU (Adamo et al. 2024). Of that 50%, more than 70% still comes from fossil fuels (Adamo et al. 2024). Recently, the International Energy Agency (IEA) highlighted industrial large-scale heat pumps (IHPs) for their promising ability to electrify industrial heating (IEA 2026). Today, buildings still drive IHP deployment, but industry heat carries significant growth potential (IEA 2026).

In the Netherlands, IHPs already contribute to industrial energy efficiency improvements. For example, the Dutch Environmental Assessment Agency (PBL) stated that this technology improves energy savings across paper and board production, chemical processes, and steam cracking (PBL 2023). IHPs could already cover around 20% of global industrial heat demand, primarily in low-to-medium-temperature processes (IEA 2026). However, integrating them into existing infrastructure is not always economically viable. Therefore, deployment requires policy support.

Economic policies can make IHPs more cost-competitive with fossil-fuel heaters, like gas boilers. Smillie et al. (2024, p. 13) argue investment tax credits and government-sponsored low-cost financing can lower IHP CAPEX. In the Netherlands, an example of such a policy is the Sustainable Energy Production and Climate Transition Incentive Scheme (SDE++). This policy provides subsidies to companies and non-profit organisations that reduce CO_2 emissions on a large scale (Netherlands Enterprise Agency 2025).

Besides subsidies, Smillie et al. (2024) also find that carbon pricing is a promising policy route. In the EU, the Emissions Trading System (ETS) already drives industrial heat electrification through carbon pricing. Under this scheme, industry actors using fossil-fuel heaters above 20 MW face rising prices for EU carbon allowances (EUAs) and a gradual phase-out of free EUA allocation (European Parliament 2003). The Carbon Border Adjustment Mechanism (CBAM) extends this carbon price to carbon-intensive imports (European Commission 2026b). Together, ETS and CBAM create favourable conditions for IHPs.

Some scholars argue that ETS does not deliver the emission reductions necessary to halt climate change because it does not sufficiently drive up carbon prices (Ervine 2018; Hintermayer 2020). As a solution, the Market Stability Reserve (MSR) was introduced in 2018 to remove a surplus of EUAs that historically kept prices low (WKR 2026). Since 2023, the MSR has been removing EUAs from the market and retaining at most 400 million allowances. Extra EUAs above that limit are invalidated each year, permanently removing them from the system and driving up the carbon price (WKR 2026). In April 2026, the European Commission proposed ending this invalidation mechanism to prepare for geopolitical instability and energy insecurity (European Parliament 2026). Climate groups and Members of the European Parliament expressed a fear that this would weaken the market (Stoefs 2026). In September 2026, EU Parliament instead proposed keeping invalidation but raising the threshold to 650 million from February 2027 to preserve a buffer while preventing excessive accumulation of EUAs (WKR 2026). The outcome of this decision will shape future scarcity under the ETS, and therefore EUA prices.

This report examines how the combination of improved IHP CAPEX and efficiency along with carbon pricing affect the supply of industrial low-to-medium-temperature heat demand between 2020 and 2050.

The main question is:

> **What is the effect of heat pump technology (CAPEX and efficiency) improvements and carbon pricing on the deployment of gas boilers and heat pumps to fulfil industrial low-to-medium temperature heat demand in the Netherlands from 2020 to 2050?**

Methodology

To model the Dutch energy system, this report used PyPSA (PyPSA Contributors 2026). Industrial gas boilers and industrial heat pumps (IHPs) were implemented as PyPSA links. Gas boilers took gas as their main input, while IHPs took electricity. Both technologies delivered industrial low-to-medium temperature process heat as their main output (Figure 1). Since gas boilers burn gas, they also had a secondary output to atmospheric CO_2, which IHPs did not have.

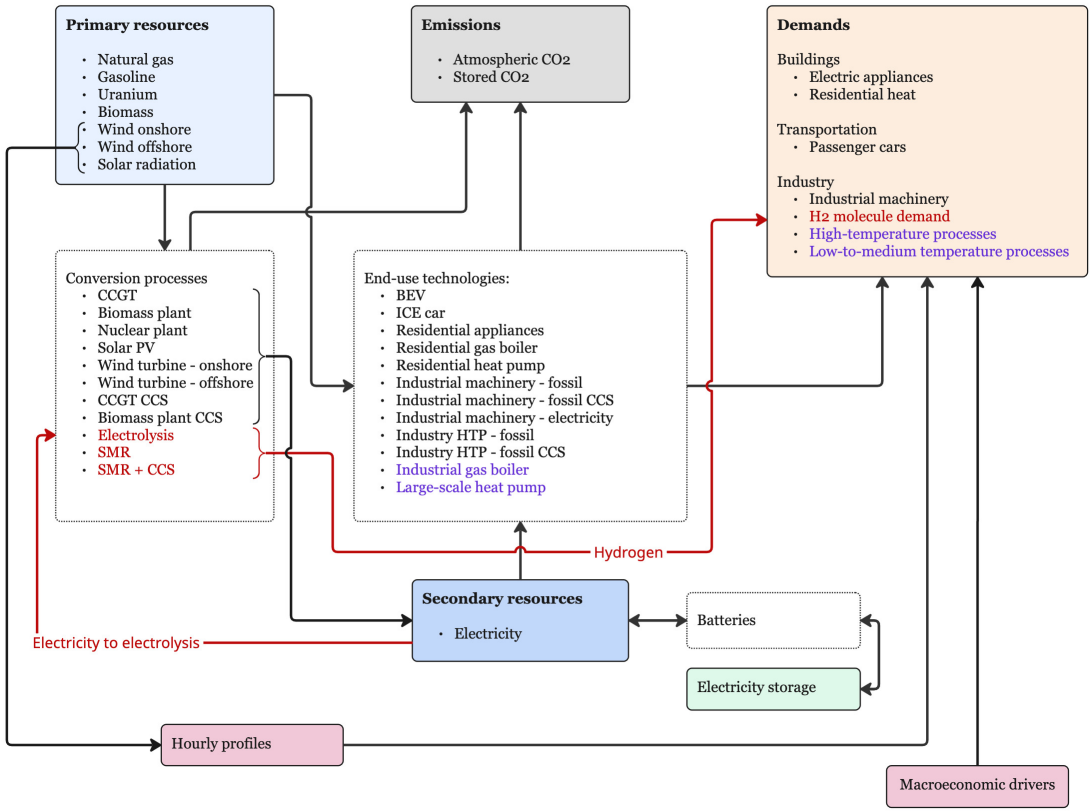


Figure 1: Diagram of the full model topology. Black text and arrows show the parts based on the course Modelling for Transitions (Dalla Longa 2026). The black arrows between components indicate inputs and outputs, but not every listed component necessarily has all of the inputs and outputs shown. Red text and arrows show the topology as implemented in the group modelling assignment that preceded this individual assignment. Purple text marks the additions made specifically for this individual assignment.

Four scenarios modelled the cost-optimal supply of heat for industrial low-to-medium-temperature processes, combining default and improved IHP technology with default and high carbon prices (Table 1). All multipliers started at 1.0 in 2020. Under default technology assumptions, IHP CAPEX and efficiency experiences $\times 1$ multipliers for all model years. Under improved technology assumptions, the CAPEX multiplier decreased by 20%, while efficiency increased by 12% in 2050 relative to the 2020 baseline. The carbon-price multiplier remained at 1.0 in the low-price scenarios and increased to 12.0 by 2050 in the high-price scenarios.

The improved-technology multipliers drew on relative CAPEX reductions and efficiency gains reported in Achhammer and Hampp (2024) and Grosse et al. (2017, p. 63). The baseline carbon price was based on the reported 2020 price from the Department for Business, Energy & Industrial Strategy (BEIS) (2019). The high-price multiplier was chosen to support substantial changes in the model solution for analysis. The final $\times 12$ multiplier increased the carbon price to 300 €/*ton*, which sits slightly above the 250 $/*ton* price projections for 2050 by the IEA

(Reid 2024). Absolute input values and model topology details are available on GitHub (Zuijderwijk 2026), with further parametrization details in Appendix B, and model assumptions in Appendix C.

Table 1: Scenario definitions and parameter multipliers for industrial large-scale heat pump (IHP) technology and carbon price.

Scenarios	**IHP tech (CAPEX & efficiency)**	**Carbon price**
Default Tech, Default Carbon Price (S1)	Default	Default
Improved Tech, Default Carbon Price (S2)	Improved	Default
Default Tech, High Carbon Price (S3)	Default	High
Improved Tech, High Carbon Price (S4)	Improved	High

Parameter assumptions	**2020**	**2030**	**2040**	**2050**
Improved large-scale heat pump CAPEX	1.00	0.93	0.87	0.80
Improved large-scale heat pump efficiency	1.00	1.04	1.08	1.12
Higher carbon price	1.00	4.00	8.00	12.00
Regular assumptions	1.00	1.00	1.00	1.00

Results

In the baseline scenario (S1), industrial heat provision shifted from gas boilers to IHPs in 2050 (Figure 2). Improved technology (S2) and high CO_2 pricing (S3) both brought this transition forward to 2040, with S3 achieving a slightly higher IHP share in 2040 than S2 (Figure 2). Combining higher carbon pricing with improved technology (S4) brought IHP deployment forward to 2030 (Figure 2). The high-carbon-price scenarios (S3 & S4) also installed more gas boiler capacity in 2020.

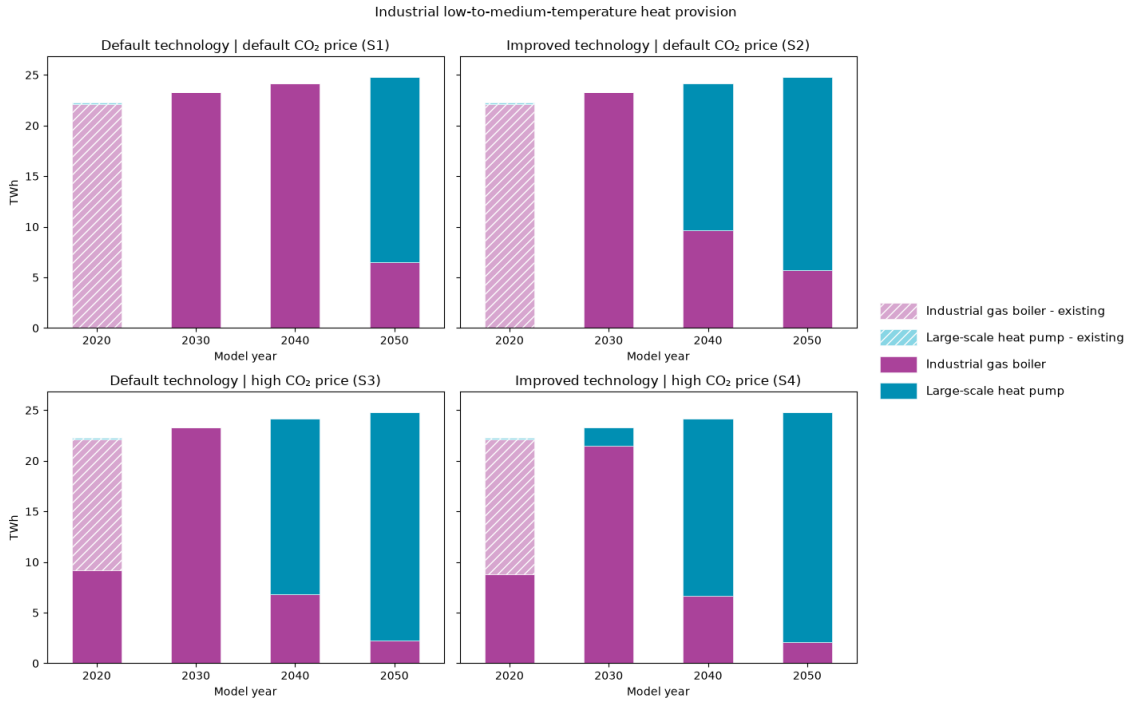


Figure 2: Heat supply from gas boilers and large-scale heat pumps across four technology and CO_2 price scenarios from 2020 to 2050. Hatched segments indicate existing assets.

The baseline scenario (S1) produced the highest total system emissions of 2027.4 $MtCO_2$, while improved IHP technology under default carbon pricing (S2) yielded a modest reduction to 1993.9 $MtCO_2$(Table 2). Higher carbon pricing (S3) drove a larger decrease to 1771.0 $MtCO_2$ (Table 2). Combining higher carbon pricing with improved technology (S4) lowered emissions further to 1763.0 $MtCO_2$ (Table 2). Gas boiler emissions declined across these scenarios as well, with carbon pricing having a more pronounced effect than improved technology (S3 & S2) (Table 2).

Table 2: Cumulative net system and direct industrial gas-boiler emissions from modelled years, 2020–2050 ($MtCO_2$).

Scenario	System	Industrial Gas boilers
Default Tech, Default Carbon Price (S1)	2027.4	137.2
Improved Tech, Default Carbon Price (S2)	1993.9	104.2
Default Tech, High Carbon Price (S3)	1771.0	94.0
Improved Tech, High Carbon Price (S4)	1763.0	89.4

Across both high carbon pricing scenarios (S3 & S4), emission reductions picked up over modelled years as the carbon price multiplier increased (Figure 3A). The increasing carbon price caused net total emission reductions of 256.4 $MtCO_2$ and 264.4 $MtCO_2$ in S3 and S4, respectively (Figure 3B). On the other hand, technology improvement alone (S2) yielded a 33.5 $MtCO_2$ reduction (Figure 3B). When carbon pricing incentivized technologies that do not emit CO_2 (S3 & S4), the direct emission reductions from industrial gas boilers were higher (43.1 & 47.7 $MtCO_2$, respectively) than with improved IHP technology alone (32.9 $MtCO_2$) (Figure 3B).

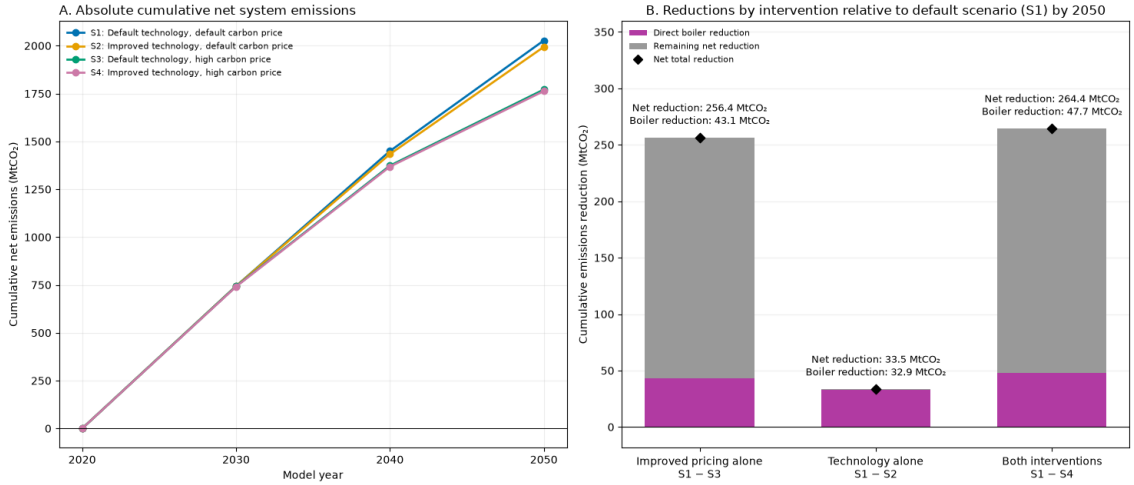


Figure 3: A. Absolute cumulative net system emissions (2020-2050) and B. cumulative emission reductions by policy relative to the default scenario (S1). On top of each bar, there is a note about the net total emission reductions that can be attributed to the bar's corresponding policy, as well as the contribution of gas boiler replacement with IHPs.

The carbon emission reductions were mirrored by an upward trend across scenarios of electricity consumption that coincided with a move away from gas-fired electricity across all scenarios (Figure 4). In high-carbon-price scenarios (S3 & S4), system-wide electricity consumption increased more than it did for scenarios with default carbon pricing (S1 & S2). In those scenarios, improved technology (S2) decoupled slightly from S1 in 2040, but converged again in 2050. Across all scenarios, electricity was mostly supplied by solar in 2050 (Figure 4B).

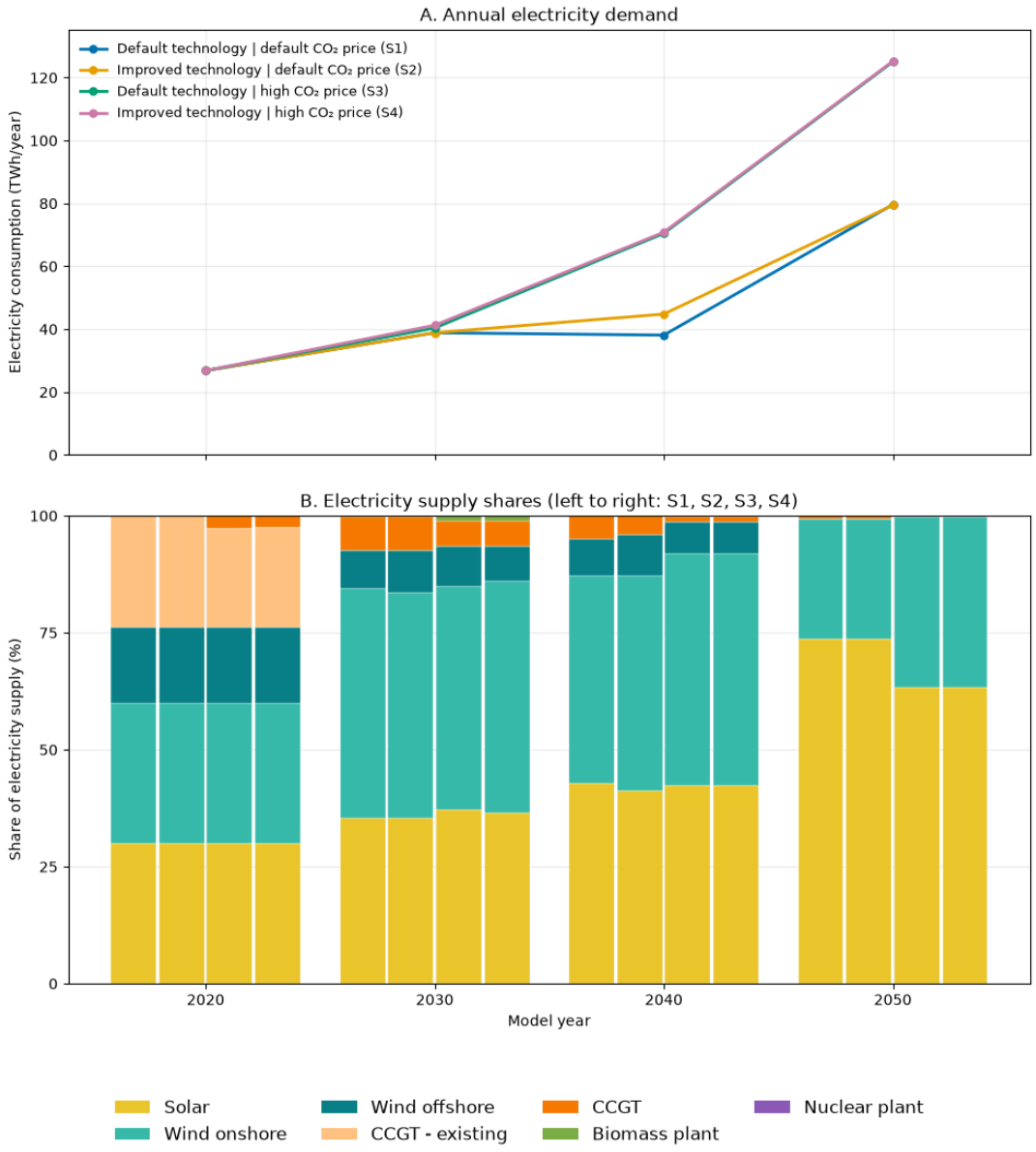


Figure 4: A. Annual system electricity demand B. Share of electricity supply by energy source.

Discussion and Conclusions

The scenario results revealed that improved technology policies electrified industrial heat on par with carbon pricing policy (Figure 2). Both improved IHP technology (S2) and high carbon price (S3) policies brought the shift from gas boilers to industrial large-scale heat pumps (IHPs) forward from 2050 to 2040 (Figure 2). When combined (S4), their synergistic effects allowed for IHP deployment to start as early as 2030 (Figure 2). However, improved technology policies lagged behind carbon pricing policy in terms of decarbonisation (Figure 3). This can be attributed to their differences in scope.

Improving IHP technology is a more targeted policy than carbon pricing. It affects only the attractiveness of IHPs. In the modelled scenarios, this increased attractiveness was responsible for earlier electrification of industrial low-to-medium-temperature heating (Figure 2). It also contributed an emissions reduction from 137.2 $MtCO_2$ (S1) to 104.2 $MtCO_2$ (S2) as a result of industrial gas boiler replacement (Table 2). However, once carbon pricing made gas boilers less

competitive, their emissions reduced further from 137.2 $MtCO_2$ to 94 $MtCO_2$. This implies that carbon pricing is a more effective policy for concurrently electrifying and decarbonising industry (Table 2).

Carbon pricing is a stronger driver of electrification and decarbonisation because it applies to all CO_2-emitting processes and technologies in the system. Where technology improvements made IHPs more attractive, carbon pricing drove two concurrent shifts: one in the cost-competitiveness of IHPs over gas boilers, and another in the move from CO_2-emitting to renewable electricity generation. This is evidenced by the relatively larger cumulative system emission reductions by 256.4 $MtCO_2$ and 264.4 $MtCO_2$ respectively in S3 and S4 (Figure 3B). For S3, 43.1 $MtCO_2$ (16.8%) of this reduction came directly from replacing industrial gas boilers with IHPs. The remaining 213.2 $MtCO_2$ (83.2%) stemmed largely from increased electrification across the rest of the system (Figure 4A). This electrification happened in residential heating, transportation, and hydrogen technologies as well as industrial heating (see Appendix D for these sectors). At the same time, electricity generation decarbonised (Figure 4B). Therefore, in high-carbon-price scenarios the system used more electricity, and more of that electricity was generated through renewable processes (Figure 4).

For policy makers aiming to electrify and decarbonise industrial heating, this suggests that economic policy support for improving heat pump costs and efficiency can produce the best results when combined with a strong carbon price signal. Government programs that lower IHP CAPEX, such as SDE++, can support heat pump adoption. However, on its own, this kind of policy achieved relatively modest emission reductions in the modelled scenarios (S2) (Figure 3). Combining carbon pricing with technology improvement (S4) reduced emissions further and brought IHP adoption for industrial heat supply forward (Figure 2 & Figure 3). Therefore, combining policies can speed up the electrification of industrial heat and drive decarbonisation.

Under the ETS scheme, this highlights the importance of the Market Stability Reserve (MSR) that stimulates the increase of the carbon price over time by invalidating the surplus of EUAs (WKR 2026). If the European Commission decides to end this invalidation mechanism, carbon pricing might weaken (Maulberger and Rathgeber 2026). In such a case, this report's S1 and S2 scenarios are likely closer to the future system-wide emissions trajectory than S3 and S4. In other words, electrification and decarbonisation of low-temperature heat would likely be less pronounced if the carbon price weakens.

These insights should come with caution as the model has a set of limitations. For example, the model ignored grid congestion, which currently forms a major barrier to electrification plans in the Netherlands (Ecorys 2024; Water & Energy Solutions 2024). As a result, the pace of electrification is likely overstated in this report. Additionally, the model assumed gas price remained constant throughout the model years for all scenarios, meaning that the model did not capture the impacts of geopolitical risks and fossil-fuel supply volatility. On top of that, the ETS 1 scheme imposes a carbon cap, which this model did not implement. The scheme also only includes industrial gas boilers with power exceeding 20 MW. This model generalised all industrial gas boilers and did not select for those specific facilities. Finally, CAPEX decreases were exogenously implemented, meaning that there was no feedback between the system and CAPEX values. For more details on model assumptions, see Appendix C.

In light of these limitations, future work should address these gaps by adding grid constraints, gas price scenarios, and a carbon cap. Further testing could focus on gas prices. This could deliver insights on energy security and system responses to fossil-fuel price volatility. Lastly, risk perception is an important factor in industrial decarbonisation. Therefore, adding an agent-based behavioural layer to the model could provide information on how firms adopt renewable technologies when influenced by economic risks.

In conclusion, the results are indicative of the relative effects of the covered policies. Carbon pricing and technology improvement have similar and synergistic effects on the pace of industrial heat electrification under the model's assumptions. However, carbon pricing has a higher decarbonisation potential by making all CO_2-emitting processes and technologies more expensive. For both electrification and decarbonisation of the industrial heat sector, IHP technology improvement policies work best as a complement to strong carbon pricing policy.

References

Achhammer, Anton and Johannes Hampp (2024). *PyPSA Technology Data: Cost and Technical Assumptions for Energy System Modeling*. https://github.com/PyPSA/technology-data. Accessed: 2026-10-21.

Adamo, Angela et al. (2024). "Regulatory framework for the promotion of heat pumps in the European Union". In: *Energy Reports* 12, pp. 5909–5930. DOI: 10.1016/j.egyr.2024.11.043. URL: https://doi.org/10.1016/j.egyr.2024.11.043.

Burghouwt, Karin (Apr. 2021). "Industriële warmtepomp begonnen aan voorzichtige opmars: Aanpak aantal knelpunten nodig voor doorbraak". In: *VEMW Nieuwsbericht*. URL: https://www.vemw.nl/nieuwsbericht/2021/04/02/Industri%C3%ABle-warmtepomp-begonnen-aan-voorzichtige-opmars.

Dalla Longa, Francesco (2026). *Lecture Presentation*. Lecture notes / Course material distributed via Canvas. Accessed: 2026-10-22. URL: https://canvas.uva.nl/courses/59604/files/15890622?module_item_id=3031478.

Department for Business, Energy & Industrial Strategy (BEIS) (2019). *Determination by the Secretary of State for Business, Energy and Industrial Strategy of the EU Emissions Trading System (EU ETS) Carbon Price under the Greenhouse Gas Emissions Trading Scheme Regulations 2012*. Tech. rep. Published 29 November 2019 for the 2020 scheme year; Accessed: 2026-10-22. London, UK: Government of the United Kingdom. URL: https://assets.publishing.service.gov.uk/media/6479e55fb32b9e000ca960cc/eu-ets-carbon-penalty-price-determination-2020.pdf.

Ecorys (Apr. 2024). *Maatschappelijke Kostprijs van Netcongestie: Eindrapport*. Commissioned by Rijksdienst voor Ondernemend Nederland (RVO) and the Ministry of Economic Affairs and Climate Policy. Ecorys. URL: https://www.ecorys.com/app/uploads/2019/02/Maatschappelijke-kostprijs-van-netcongestie-2024.pdf.

Energie Beheer Nederland (EBN) (Jan. 2026). *Infographic: Energy in Numbers 2026*. Tech. rep. Utrecht, The Netherlands: Energie Beheer Nederland (EBN). URL: https://www.ebn.nl/en/facts-figures/knowledge-base/infographic-2026/.

Ervine, Kate (2018). "How Low Can It Go? Analysing the Political Economy of Carbon Market Design and Low Carbon Prices". In: *New Political Economy* 23.6, pp. 690–710. DOI: 10.1080/13563467.2018.1384454. URL: https://doi.org/10.1080/13563467.2018.1384454.

European Commission (Sept. 2, 2026a). *ETS2: buildings, road transport and additional sectors*. Last updated on 2 September 2026. URL: https://climate.ec.europa.eu/areas-action/carbon-markets/ets2-buildings-road-transport-and-additional-sectors_en (visited on 10/24/2026).

— (2026b). *Carbon Border Adjustment Mechanism (CBAM)*. Directorate-General for Taxation and Customs Union. URL: https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en (visited on 10/24/2026).

European Hydrogen Observatory (2026). *Hydrogen Demand*. Clean Hydrogen Partnership. URL: https://observatory.clean-hydrogen.europa.eu/hydrogen-landscape/end-use/hydrogen-demand (visited on 10/24/2026).

European Parliament (Oct. 13, 2003). *Directive 2003/87/EC of the European Parliament and of the Council of 13 October 2003 establishing a system for greenhouse gas emission allowance trading within the Union and amending Council Directive 96/61/EC*. EUR-Lex. Consolidated version of 1 March 2024. URL: https://eur-lex.europa.eu/eli/dir/2003/87/2024-03-01 (visited on 10/24/2026).

— (2026). *EU emissions trading system (ETS): Amending the market stability reserve*. European Parliament. URL: https://www.europarl.europa.eu/news/nl/press-room/20260911IPR47449/eu-emissions-trading-system-ets-amending-the-market-stability-reserve (visited on 10/24/2026).

Grosse, Robert et al. (2017). *Long term (2050) projections of techno-economic performance of large-scale heating and cooling in the EU*. EUR 28859 EN. Luxembourg: Publications Office of the European Union. DOI: 10.2760/24422. URL: https://publications.jrc.ec.europa.eu/repository/handle/JRC109006.

Hintermayer, Martin (2020). "A carbon price floor in the reformed EU ETS: Design matters!" In: *Energy Policy* 147, 111905. DOI: 10.1016/j.enpol.2020.111905. URL: https://doi.org/10.1016/j.enpol.2020.111905.

IEA (2019). *The Future of Hydrogen: Assumptions Annex*. Report Annex. Revised version, December 2020. Paris, France: International Energy Agency (IEA). URL: https://iea.blob.core.windows.net/assets/29b027e5-fefc-47df-aed0-456b1bb38844/IEA-The-Future-of-Hydrogen-Assumptions-Annex_CORR.pdf.

— (2021). *Global Hydrogen Review 2021: Assumptions Annex*. Assumptions Annex. Paris, France: International Energy Agency (IEA) and IEA Greenhouse Gas R&D Programme (IEAGHG). URL: https://iea.blob.core.windows.net/assets/2ceb17b8-474f-4154-aab5-4d898f735c17/IEAGHRassumptions_final.pdf.

— (2026). *Heat Pump Monitor 2026: Key Findings*. Tech. rep. Accessed: 2026-10-22. Paris, France: International Energy Agency. URL: https://www.iea.org/reports/heat-pump-monitor-2026/key-findings.

IPCC WG III Report (2022). *Climate Change 2022: Mitigation of Climate Change*. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, UK and New York, NY, USA: Cambridge University Press. DOI: 10.1017/9781009157926. URL: https://www.ipcc.ch/report/ar6/wg3/.

Maulberger, Andreas and Andreas W. Rathgeber (2026). "Disentangling supply and demand shocks in the EU ETS – Before and after the introduction of the Market Stability Reserve". In: *Resource and Energy Economics* 87, 101575. DOI: 10.1016/j.reseneeco.2026.101575. URL: https://doi.org/10.1016/j.reseneeco.2026.101575.

Netherlands Enterprise Agency (2025). *Stimulation of sustainable energy production and climate transition (SDE++)*. Page content last checked on 7 August 2025. URL: https://english.rvo.nl/subsidies-financiering/sde (visited on 10/24/2026).

PBL (2023). *Energy Efficiency Options in the Dutch Manufacturing Industry*. Report 5256. Manufacturing Industry Decarbonisation Data Exchange Network (MIDDEN); Accessed: 2026-10-22. The Hague, Netherlands: PBL Netherlands Environmental Assessment Agency. URL: https://www.pbl.nl/en/publications/energy-efficiency-options-in-the-dutch-manufacturing-industry.

Pixabay (2026). *CPU Heatsink Fan*. Canva. URL: https://www.canva.com/graphics/MAEFg35ZSc0/ (visited on 10/24/2026).

PyPSA Contributors (2026). *PyPSA: Python for Power System Analysis — Documentation*. Accessed 23 October 2026. URL: https://docs.pypsa.org/latest/ (visited on 10/23/2026).

Reid, Andrew (Nov. 7, 2024). *Fact Sheet: Carbon Pricing in the EU*. Fact sheet. Institute for Energy Economics and Financial Analysis. URL: https://ieefa.org/resources/fact-sheet-carbon-pricing-eu.

Smillie, Sean et al. (Oct. 2024). *Decarbonizing Industrial Heat: Measuring Economic Potential and Policy Mechanisms*. Tech. rep. Prepared for the Center for Applied Environmental Law and Policy (CAELP). Energy and Environmental Economics, Inc. (E3). URL: https://www.ethree.com/wp-content/uploads/2024/10/CAELP-E3-Industrial-Electrification-Report.pdf.

Stoefs, Wijnand (June 4, 2026). *Shooting blanks: Why 'increasing the firepower' of the Market Stability Reserve misses the target*. Briefing. Carbon Market Watch. URL: https://carbonmarketwatch.org/publications/shooting-blanks-why-increasing-the-firepower-of-the-market-stability-reserve-misses-the-target/.

Water & Energy Solutions (Nov. 2024). *Roadmap Energietransitie Havenindustrie Amsterdam (REHA)*. Port of Amsterdam and Gemeente Amsterdam. URL: https://openresearch.amsterdam/image/2024/11/18/roadmap_energietransitie_havenindustrie_amsterdam_reha.pdf.

WKR (July 3, 2026). *Factsheet MSR*. URL: https://www.wkr.nl/documenten/2026/07/03/factsheet-msr (visited on 10/24/2026).

Zuijderwijk, Noah (Oct. 2026). *PyPSA_Dutch_EnergySystem: FinalModel/input_databases/default*. URL: https://github.com/nzuijder/PyPSA_Dutch_EnergySystem/tree/main/FinalModel/input_databases/default (visited on 10/22/2026).

A Use of LLMs

I have read and understood the UvA rules regarding fraud and plagiarism. I declare that this written paper is entirely my own work, that I have carefully and correctly indicated all the sources I have used, and that I have cited according to the rules. I have not previously submitted this paper, in this or an amended version, for another subject or as part of another paper. I have used AI tools only for the purposes cited below:

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I find that LLMs can get stuck in a certain misunderstanding of my prompt. In those cases, I switched between LLMs, which often helped resolve the misunderstanding. This explains why I report on chats from Claude, Gemini and UvA AI Chat.

B Parametrization for model inputs

In the preceding group modelling assignment, the group implemented a scenario with a modified gas price. A start year of 2025 was therefore chosen to avoid anomalous COVID-19-related gas prices in 2020 for model calibration. For this report, the start year was shifted back from 2025 to 2020 to make all intervals 10 years long. Consequently, new data for hydrogen technologies in 2020 had to be sourced. Data was found only for 2019 and 2021 (IEA 2019; IEA 2021). The 2019 and 2021 values were averaged to generate the inputs for the dynamic parameters of hydrogen technologies in 2020 (Table 3). Additionally, the updated hydrogen molecule demand was based on 2022, which was the earliest available data from the European Hydrogen Observatory (2026) (Table 4). To model the scenarios in this report, start values for industrial heat demand and existing heating technology power were identified and implemented (Table 4).

Table 3: Hydrogen technologies parametrization (2020).

Parameter	SMR	SMR+CCS	Electro.	Unit	Source / Notes
CAPEX	845	1575	1325	€/kW	Updated to 2020 start year from averaged 2019 and 2021 data (IEA 2019; IEA 2021)
FOM	39.72	55.13	39.75	€/kW/yr	Same source as CAPEX
VOM	0	0.0078	0	€/kWh	Same source as CAPEX
Efficiency	0.76	0.69	0.64	–	Same source as CAPEX
Lifetime	30	30	30	Years	Same source as CAPEX
Emissions	0.20	0.01	–	tCO_2/MWh	Same source as CAPEX
Captured CO_2	–	0.19	–	tCO_2/MWh	Same source as CAPEX

Table 4: Existing heating technologies and demand parametrization (2020).

Parameter	Value	Unit	Source / Notes
Installed heat pump power	50	MW	Estimated power in 2020 based on (Burghouwt 2021)
Installed gas boiler power	4200	MW	Existing industrial power in 2020
HTP heat demand	39583075.56	MWh	64% of total heat demand derived from final course model demand value (Energie Beheer Nederland (EBN) 2026)
LTP heat demand	22265480.00	MWh	36% of total heat demand derived from final course model demand value (Energie Beheer Nederland (EBN) 2026)
Updated H2 molecule demand	23681295.40	MWh	Converted ton/year to MWh/year using hydrogen's LHV of about 33.33 MWh per metric ton, based on earliest available data (from 2022) (European Hydrogen Observatory 2026)

For technology financial data, no specific sources were found that corroborated the CAPEX of 400 €/kW and efficiency of 2 given in the assignment description. Therefore, CAPEX and efficiency percentage changes between 2020 and 2050 were noted. Across sources, the IHP CAPEX falls by 18-20%, while HP efficiency increases by 12% (Table 5).

Table 5: Heat pump baseline assumptions and supporting trends (2020–2050).

Parameter	2020	2050	Unit	Source / Notes
Reported CAPEX	660	540	€/kW	Electric large-scale heat pumps; approximately 18% decrease. Converted from 0.66 and 0.54 M€/MW (Grosse et al. 2017, p. 63)
Reported CAPEX	1103.06	886.30	€/kW	Industrial medium-temperature heat pumps; approximately 20% decrease (Achhammer and Hampp 2024)
Reported efficiency	2.55	2.85	–	Approximately 12% increase (Achhammer and Hampp 2024)
Assumed CAPEX change	–	-18 to -20	%	Relative change from 2020, based on the reported CAPEX trends above
Assumed efficiency change	–	+12	%	Relative change from 2020, based on the reported efficiency trend above

C Model assumptions

These model assumptions were made to fit the scope of the individual modelling assignment. They toe the balance between technical rigour, model running time, and pragmatism with regards to the one-week timeline for this assignment. For details on parameter values, see (Zuijderwijk 2026):

1. Model resolution is set at $t_res = 7hrs$ to capture daily solar cycles, as well as wind patterns, while keeping running time reasonable
2. Industrial gas boiler and large-scale heat pump end-use technologies are the only technologies that supply industrial_ltp demand
3. All industrial demand profiles industrial_machinery industrial_htp industrial_ltp industrial_hydrogen are constant, built on the assumption that industrial demand is relatively constant
4. In default scenarios, the IHP technology CAPEX and efficiency, and the carbon price maintain their 2020 values up to 2050 to observe the contrast between policy scenarios and the default scenario
5. The gas price remains constant throughout the model years for all scenarios, meaning that industrial gas boilers do not experience changes in fuel costs
6. The carbon price is applied to all industrial gas boilers in the model. However, under ETS 1, only facilities with power exceeding 20 MW are included. However, an updated ETS scheme (ETS 2) will go into effect in 2028. This updated scheme will include process heat in smaller-scale industries (European Commission 2026a).
7. Earliest available 2022 data on hydrogen molecule demand from European Hydrogen Observatory (2026) is assumed to sufficiently represent 2020 hydrogen molecule demand
8. Electrolysis, SMR, and SMR with CCS create hydrogen that is not linked to any end-use technologies and hydrogen is not modelled as a secondary resource. Instead, hydrogen supplies a generalised industrial H2 molecule demand that has 3% overlap with industrial heat demand (European Hydrogen Observatory 2026)
9. Grid congestion is ignored as it did not fit within the scope of this report
10. No carbon cap is implemented as this report focused on carbon price

D Supplemental figure

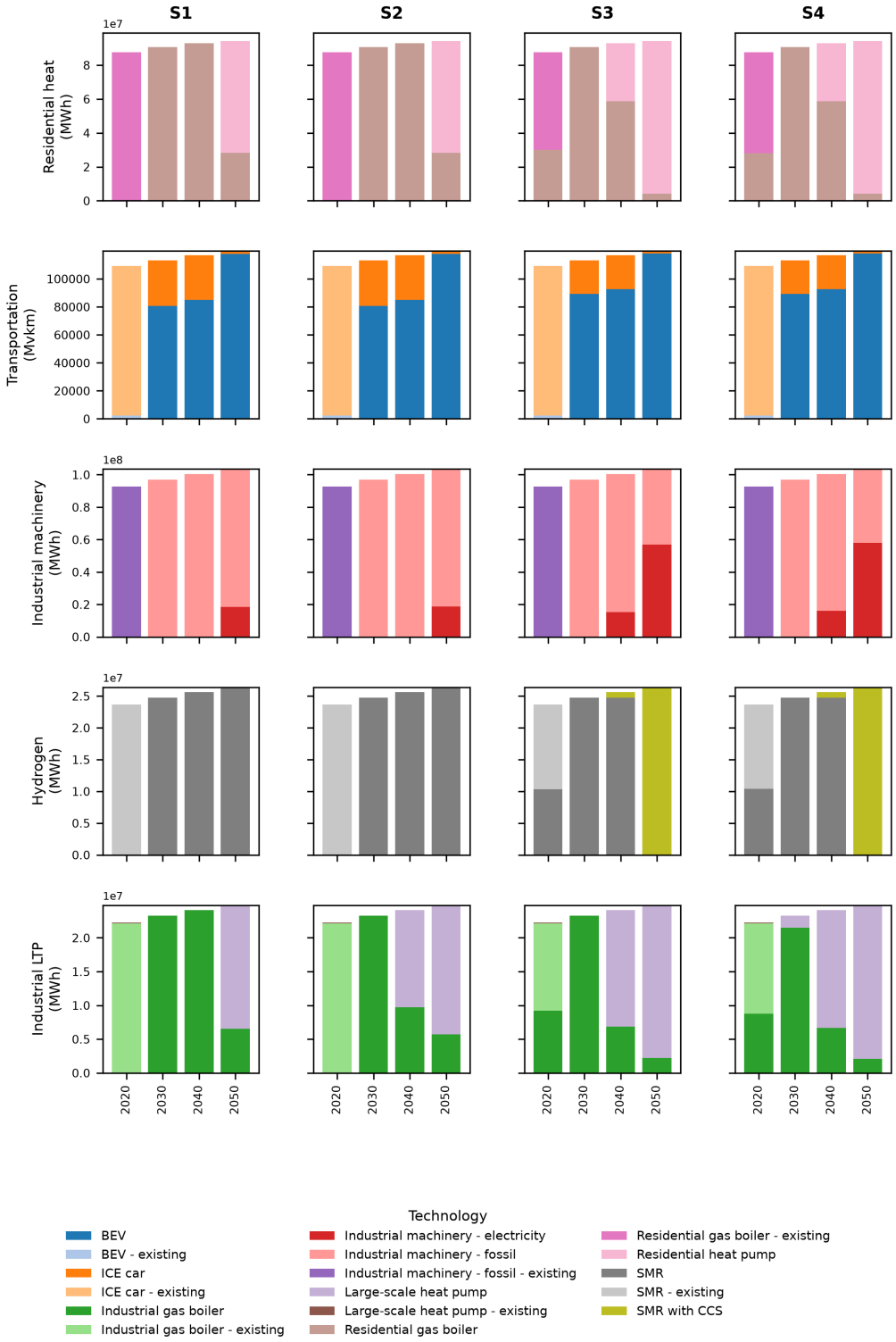


Figure 5: A system perspective on energy balances across sectors per scenario.